JEEVAN SAI KANAPARTHI

Software Application Engineer 3 (Mid Level) | Software Engineer | Chennai, India
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Notice Period: 1 Month

Technical Skills Summary

- **Cloud & Infrastructure:** AWS (S3, DynamoDB, AWS Lambda, SQS, SNS, CloudWatch, EKS, RDS, IAM, VPC, Load Balancers), Terraform, Ansible, AWS CloudFormation
- **DevOps & CI/CD:** CI/CD (GitHub Actions), Docker, Kubernetes, Maven
- **Programming & Scripting:** Python, TypeScript, JavaScript, Shell Scripting, PHP.
- **Databases & Data:** DynamoDB (GSI's), Snowflake SQL, MySQL, SQL, Freemarker Templates.
- **API & Web Development:** REST API, Slack API, HTML, CSS, Angular, Fast API
- **Fundamentals:** Data Structures & Algorithms, Linux Administration.
- **Version Control:** Git, GitHub, Bitbucket, AWS CodeCommit.
- **Artificial Intelligence:** Amazon Bedrock, Agentic AI, Model Context Protocol (MCP)

Experience

Software Application Engineer III, Amazon — Chennai
September 2025 – Present I **Notice Period: 1 Month**
Key Focus: Distributed Systems, Operational Excellence, Automation & Production Reliability

- **COptimus — AI Code Review Helper** – Built *COptimus*, an AI-powered code review assistant leveraging internal automation tools (*kiro-cli*, *MCP*) to pre-analyze change requests with team-specific context. Trained on 100+ historical CRs to learn file relationships and generate prioritized findings (P1–P3) across 14 verification dimensions. Validated on live team reviews, producing context-aware insights referencing configuration inheritance and field-level consistency. Distributed internally for team-wide adoption through MCP integration.
- **Google IAP Marketplace Expansion (MPE)** – Contributed to enabling in-app billing for digital content across six new international marketplaces. Delivered end-to-end configuration, onboarding, and validation workflows, achieving full production launch in Japan and UK. Accelerated rollout for four additional regions, improving global billing coverage and customer accessibility.
- **Alternate Billing Enablement (US)** – Extended alternate billing support to the US marketplace, aligning parity with EU regions and enhancing payment flexibility through backend configuration and event routing improvements.
- Designed and developed **HawkEye**, a scalable **transaction validation and observability platform** for automating financial reconciliation workflows across digital content marketplaces. Engineered a centralized **REST API** to consolidate transaction events, source records, and expected accounting outputs into a unified validation pipeline, enabling one-step data retrieval with filtering by **business line, country, and transaction type** along with pagination support for large datasets. Eliminated a highly manual investigation process that previously required navigation across **5+ distributed systems** and analysis of **11 database tables**, significantly reducing operational overhead and improving investigation efficiency. Designed the platform to operate at scale across **500+ transaction types**, **5 business lines**, and **8 global marketplaces**, with robust error handling, high test coverage, and production-grade reliability mechanisms to ensure financial data integrity.
- Engineered a critical migration of UI localization resources from legacy storage to a high-performance database architecture. Took **end-to-end ownership** of the redesign after identifying limitations in the initial approach, delivering a **single-script deployment process** with **zero downtime** across global production sites.
- Improved microservice reliability and observability by resolving **35+ production issues** and architectural risks in high-traffic distributed systems. Reduced false-positive alert noise from **90% to**

0.03% for critical enrollment services, effectively eliminating **alarm fatigue** and improving operational response efficiency.
- Developed a custom automation suite using **JavaScript** and **Slack API** integrations to monitor and triage stale support tickets across multiple engineering resolver groups. Automated high-severity reporting workflows, significantly reducing manual overhead and accelerating issue response times.
- Resolved complex UI architectural issues involving state retention and search workflows, improving operational efficiency for publisher and catalog-management teams handling global content operations.
- Championed **code quality**, peer review standards, and operational excellence initiatives, successfully delivering **38+ technical resolutions** and mitigating **15 architectural risks** within the first year.

Software Engineer, NielsenIQ India — Chennai

February 2022 – August 2025

Key Focus: Python Automation, Containerization, System Optimization & Data Processing

- Developed **Python automation scripts** optimizing **container management** and **orchestration** with **Docker** and **Kubernetes**, improving deployment efficiency.
- Applied **data structures** and **algorithms** in **Python** to create efficient **system-level automation**, streamlining repetitive tasks and enhancing performance by **30%**.
- Spearheaded **Docker containerization**, streamlining deployments and eliminating errors.
- Automated complex **query generation** using **Snowflake SQL**, **Freemarker Templates**, and **window functions**, improving **reporting accuracy** by **45%**.
- Mentored **junior engineers** in **Code best practices**, boosting team productivity by **15%**.
- Optimized **build** and **dependency management** workflows using **Maven**, improving **build consistency**.
- Upskilled in **cloud technologies**, optimizing **system efficiency** and reducing **operational costs** by **25%**.
- Focused on **Docker containerization**, ensuring efficient **deployments** across all environments.
- Expanded expertise in **data structures**, **algorithms**, and **system optimization** for efficient **automation** and **infrastructure management**.
- Utilized **Python** for **database interactions**, enabling efficient **data manipulation** and **integration** with **Snowflake SQL** and other **databases**, enhancing **data processing workflows**.

Education

- B.Tech from SRM University, Computer Science Engineering, Amaravathi, April 2018 - April 2022, **CGPA: 9.31**

Certifications

- Monitoring **AWS** with **CloudWatch** - LinkedIn Learning
- Advanced **Python**: Working with **Databases**
- Learning Amazon Web Services **Lambda**
- Using Python for **Automation**
- **Model Context Protocol (MCP)**: Hands-On with Agentic AI
- **Angular** Essential Training
- **Agentic AI** Fundamentals: Architectures, Frameworks, and Applications
- Building AI Applications with **Amazon Bedrock**
- Build **REST APIs** with **FastAPI**
- Introduction to **AWS CloudFormation**
- **DevOps** Beginners to Advanced with Projects (2023) - Udemy

Projects

- **COptimus — AI Code Review Helper (Amazon) (3 Months)**
  Built **COptimus**, an AI-powered code review assistant leveraging internal automation tools (kiro-cli, MCP) to pre-analyze change requests with team-specific context. Trained on **100+ historical CRs** to learn file relationships and generate prioritized findings (P1–P3) across **14 verification dimensions**. Validated on live team reviews, producing **context-aware insights** referencing configuration inheritance and field-level consistency. **Distributed internally** for team-wide adoption through MCP integration.
- **Google IAP Marketplace Expansion (MPE) (Amazon) (3 Months)**
  Contributed to enabling **in-app billing** for digital content across **six new international marketplaces**. Delivered end-to-end configuration, onboarding, and validation workflows, achieving **full production launch in Japan and UAT completion in the UK**. Accelerated rollout for four additional regions, improving **global billing coverage** and **customer accessibility**.
- **Alternate Billing Enablement (US) (Amazon) (3 Months)**
  Extended **alternate billing support** to the **US marketplace**, aligning parity with EU regions and enhancing **payment flexibility** through backend configuration and event routing improvements.
- **HawkEye: Distributed Transaction Validation and Observability Platform (Amazon) (3 Months)**
  Designed and developed HawkEye, a scalable platform for automating financial reconciliation workflows across digital content marketplaces. The data pipeline utilizes **AWS S3** for transactional data storage, integrating **AWS SQS** and **AWS SNS** for robust asynchronous event handling and triggering downstream processing. The backend was built using **TypeScript** and engineered a centralized **REST API** to consolidate transaction events and source records. This platform leveraged **AWS** services including **DynamoDB**, utilizing **Global Secondary Indexes (GSI's)** to enable **cost-optimized and performant querying** for filtering and data retrieval. **AWS CloudWatch** was implemented for real-time monitoring and observability, alongside mechanisms for **proper security and authentication**. The platform operates at scale across **500+ transaction types**, ensuring production-grade reliability and data integrity.
- **Microservice Reliability, Zero-Downtime Migration, and Automation Suite Development (Amazon) (6 months)**
  Led a critical **zero-downtime migration** of UI localization resources to a high-performance database architecture, delivering a **single-script deployment process** across global production sites. Enhanced microservice reliability and observability by resolving **35+ production issues** and architecting solutions that reduced false-positive alert noise from **90% to 0.03%**. Developed a custom automation suite using **JavaScript** and **Slack API** integrations for monitoring and triaging stale support tickets, accelerating high-severity issue response times. Championed operational excellence initiatives, mitigating **15 architectural risks** and delivering **38+ technical resolutions**.
- **Real-Time Data Aggregation and Filtering System Using Snowflake SQL – NielsenIQ (3+ Years)**
  Designed and implemented a real-time data aggregation and filtering system using advanced Snowflake SQL queries and data aggregation techniques, boosting data processing efficiency by 20%. Leveraged Python to optimize query performance, integrate databases for seamless data access and system functionality. Utilized data structures to streamline workflows, enhancing overall performance. For additional details on tools and impact, please refer to the "Software Engineer" role in the Experience section.
- **Containerized Doctor-Patient Interaction and Appointment System (Personal Project):**
  Developed a containerized Doctor-Patient Interaction and Appointment System, an online portal designed to streamline communication between doctors and patients while enhancing operational efficiency. The platform enables patients to request appointments and doctors to manage schedules and generate receipts seamlessly, fostering direct and effective interaction. The portal was built using HTML, CSS, JavaScript, PHP, and MySQL for robust front-end, back-end, and database management, ensuring a user-friendly experience. To optimize scalability and deployment, the application was containerized using Docker and Docker Compose, with multiple Docker images sourced from Docker Hub to efficiently set up and manage the containerized environment.